# Jiarui Li

PhD Student · Computer Science and Engineering

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## Research Interests

Robotics and Embodied AI Robustness, Cyber Physical Systems Security

## Education

**University of Michigan** — *Ann Arbor, MI, USA*
Ph.D. in Computer Science and Engineering — *Sep. 2024 – Present*
- Advisor: Prof. Z. Morley Mao

**Carnegie Mellon University** — *Pittsburgh, PA, USA*
M.S. in Artificial Intelligence Engineering – Information Security — *Sep. 2022 – May 2024*
- Advisor: Prof. Carlee Joe-Wong

**The Chinese University of Hong Kong, Shenzhen** — *Shenzhen, China*
B.Eng. in Computer Science and Engineering — *Sep. 2018 – Jun. 2022*

## Professional Experience

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| 2024-2026 | **Graduate Student Research Assistant**, Computer Science and Engineering, University of Michigan |
| 2023-2024 | **Graduate Research Assistant**, Electrical and Computer Engineering, Carnegie Mellon University |
| 2022-2023 | **Graduate Teaching Assistant**, Information Networking Institute, Carnegie Mellon University |
| 2021-2022 | **NLP Research Intern**, Tencent AI Lab |
| 2020-2021 | **Programming Lead**, AgriFuture |

## Publications

### Published

**Jiarui Li**, Joseph Brewington, Q. Zhang, and Z. Morley Mao. 2026. Banshee: Target Switch Attacks on Gimbal-Stabilized Visual Tracking Systems via Acoustic Injection. *47th IEEE Symposium on Security and Privacy (IEEE S&P).*

Yifan Zhang, Yuren Wang, Yunta Hsieh, Xin Wang, Ping Zhang, Ziyi Yang, Jianing Ma, Zesen Zhao, Boyuan Zheng, Hei Ting Una Chan, **Jiarui Li**, Xueshen Liu, Kunxiao Gao, Ruiyao Liu, Jingxuan Zhang, Junchen Li, Zhongwei Wan, Ziheng Zhang, Jing Xiong, Shatong Zhu, Hangrui Cao, and Hui Shen. "Speculative Decoding for Multimodal Models: A Survey." *Preprints*, 2026.

**Jiarui Li**, Joseph Brewington, Q. Zhang, and Z. Morley Mao. 2025. WIP: Hijacking Attacks on UAV Follow-Me Systems in Realistic Scenarios. *3rd USENIX Symposium on Vehicle Security and Privacy (VehicleSec).*

**Jiarui Li**, Y. Yuan, and Z. Zhang. 2024. Enhancing LLM Factual Accuracy with RAG to Counter Hallucinations: A Case Study on Domain-Specific Queries in Private Knowledge Bases. *arXiv preprint*.

P. Sharma, **Jiarui Li**, and G. Joshi. 2024. On Improved Distributed Random Reshuffling Over Networks. *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP).*

T. Kim, **Jiarui Li**, N. Madaan, S. Singh, and C. Joe-Wong. 2023. Adversarial Robustness Unhardening via Backdoor Attacks in Federated Learning. *NeurIPS Workshop on Backdoors in Deep Learning.*

W. Jiao, Z. Tu, **Jiarui Li**, W. Wang, J. Tse Huang, and S. Shi. 2022. Tencent's Multilingual Machine Translation System for WMT22 Large-Scale African Languages. *Proceedings of the Seventh Conference on Machine Translation.*

### In Review

Zesen Zhao, Joseph Brewington, Minkyoung Cho, Ruiyang Zhu, Haoran Zhang, Hui Shen, Boyuan Zheng, **Jiarui Li**, Xueshen Liu, Fan Bai, Shuqing Zeng, and Z. Morley Mao. 2026. Synergy in Action: Benchmarking Collaborative Coordination in Multi-Arm Furniture Assembly. Submitted to the *19th European Conference on Computer Vision (ECCV).*

## Research Experience

**Robustness of ML/AI-Enabled Cyber-Physical Systems Perception and Control** *Ann Arbor, MI*

ADVISOR: DR. Z. MORLEY MAO *Sep. 2024 - Present*

- **Autonomous Mobile Systems** [opensource]: Designed a robotic perception robustness framework that models sensor-to-actuator dynamics, maintains a deep learning visual object tracker, and optimizes physical control inputs under motion, latency, and system uncertainty. Validated the approach using PX4/Gazebo simulation and real-world commercial UAV experiments.
- **Collaborative Robot Learning for Multi-Arm Manipulation**: Contributed to a large-scale simulation, dataset, and benchmark for evaluating VLA policies on collaborative dual-arm manipulation, spatial alignment, and long-horizon robotic assembly.
- **End-to-End Autonomous Driving Agents**: Proposed a red teaming approach to evaluate the robustness of end-to-end Vision-Language-Action (VLA) models under safety-critical edge cases.

**Agentic LLM Robustness** *Pittsburgh, PA*

SUPERVISOR: DR. GRAHAM NEUBIG *Sep. 2023 - May. 2024*

- **Retrieval-Augmented Generation** [opensource]: Designed an end-to-end retrieval-augmented generation system for answering domain-specific and time-sensitive questions over a private knowledge base reducing hallucination.

**Machine Learning in Distributed Systems** *Pittsburgh, PA*

ADVISOR: DR. CARLEE JOE-WONG, DR. GAURI JOSHI *Sep. 2022 - May. 2024*

- **Adversarial Federated Learning**: Developed a train-time backdoor attack in which malicious federated clients selectively undermine robustness gained through adversarial training while preserving accuracy on benign inputs.
- **Convergence Analysis**: Validated an improved convergence guarantees for distributed random-reshuffling algorithms in strongly convex and nonconvex optimization, demonstrating additional speedup as the number of networked agents increases

**Multilingual Natural Language Processing** *Shenzhen, Guangdong*

MENTOR: DR. WENXIANG JIAO *Sep. 2021 - Sep. 2022*

- **Machine Translation for Low-Resource Languages**: Contributed to a large-scale multilingual neural machine translation system covering 24 African languages and 100 evaluation directions.

**Edge-to-Cloud IoT System** *Shenzhen, Guangdong*

SUPERVISOR: DR. YEH-CHING CHUNG *Sep. 2020 - Sep. 2021*

- **Real-time Agriculture System**: Designed an agricultural IoT monitoring system integrating embedded sensors, wireless communication, cloud infrastructure, and real-time data collection.

## Presentations

Summer 2026. *Banshee: Target Switch Attacks on Gimbal-Stabilized Visual Tracking Systems via Acoustic Injection*. Conference Talk, *47th IEEE Symposium on Security and Privacy (IEEE S&P).*.

Spring 2026. *Banshee: Target Switch Attacks on Gimbal-Stabilized Visual Tracking Systems via Acoustic Injection*. Invited Talk, NSF Athena AI Institute Seminar.

Spring 2026. *From Physical Signals to System Security*. Guest Lecture, Introduction to Computer Security, The University of Michigan.

Summer 2025. *WIP: Hijacking Attacks on UAV Follow-Me Systems in Realistic Scenarios*. Poster, *3rd USENIX Symposium on Vehicle Security and Privacy*.

## Awards

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| 2022 | **1st Place, Constrained Track**, WMT22 Large-Scale Machine Translation |
| 2021 | **Bronze Medal**, “Internet+” Innovation and Entrepreneurship Competition |
| 2020 | **Diligentia College Scholarship**, The Chinese University of Hong Kong, Shenzhen |
| 2019 | **”Liyun” Scholarship**, The Chinese University of Hong Kong, Shenzhen |
| 2018 | **”Liyun” Scholarship**, The Chinese University of Hong Kong, Shenzhen |

## Teaching Experience

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| Spr 2024 | **Introduction to Computer Systems**, Teaching Assistant, Carnegie Mellon University |
| F 2023 | **Introduction to Computer Systems**, Teaching Assistant, Carnegie Mellon University |
| Sum 2023 | **Mathematical Foundations for AI**, Teaching Assistant, Carnegie Mellon University |
| Spr 2023 | **Adversarial Machine Learning**, Teaching Assistant, Carnegie Mellon University |